
The Reproducibility of Adaptation in the Light of Experimental Evolution with Whole Genome Sequencing

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Abstract

A key question in evolutionary biology is the reproducibility of adaptation. This question can now be quantitatively analyzed using experimental evolution coupled to whole genome sequencing (WGS). With complete sequence data, one can assess convergence among replicate populations. In turn, convergence reflects the action of natural selection and also the breadth of the field of possible adaptive solutions. That is, it provides insight into how many genetic solutions or adaptive paths may lead to adaptation in a given environment. Convergence is both a property of an adaptive landscape and, reciprocally, a tool to study that landscape. In this chapter we present the links between convergence and the properties of adaptive landscapes with respect to two types of microbial experimental evolution. The first tries to reconstruct a full adaptive landscape using a handful of carefully identified mutations (the reductionist approach), while the second uses WGS of replicate experiments to infer properties of the adaptive landscape. Reductionist approaches have highlighted the importance of epistasis in shaping the adaptive landscape, but have also uncovered a wide diversity of landscape architectures. The WGS approach has uncovered a very high diversity of beneficial mutations that affect a limited set of genes or functions and also suggests

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some shortcomings of the reductionist approach. We conclude that convergence may be better defined at an integrated level, such as the genic level or even at a phenotypic level, and that integrated mechanistic models derived from systems biology may offer an interesting perspective for the analysis of convergence at all levels.

Keywords

Genomics • Evolution • Convergence • Adaptive landscape

11.1 Introduction

A key question in evolutionary biology is the reproducibility of adaptation. The importance of this question initially applied to its existentialist dimension, as it is directly linked to the likelihood of our own existence and also to the plausibility of the biological world. Stephen J. Gould wrote about assessing the reproducibility of adaptation: "...call this experiment 'replaying life's tape'. You press the rewind button and, making sure you thoroughly erase everything that actually happened, go back to any time and place in the past – say, to the seas of the Burgess Shale. Then let the tape run again and see if the repetition looks at all like the original. If each replay strongly resembles life's actual pathway, then we must conclude that what really happened pretty much had to happen. But suppose that the experimental versions all yield sensible results strikingly different from the actual history of life? What could we then say about the predictability of self-conscious intelligence? Or of mammals? Or of life on land? Or simply of multicellular persistence for 600 million difficult years?" (Gould 1989). This proposition by Gould is not just a *Gedankenexperiment*; it can be quantitatively addressed in the laboratory using experimental evolution coupled to whole genome sequencing (WGS). However, in the context of experimental evolution the question is no longer about the emergence of self-conscious organisms but more about the shape of an adaptive landscape and the stochastic nature of adaptation on this landscape.

11.2 Convergence and Parallelism

Convergence and parallelism are terms that characterize the reproducibility of adaptation (Haas and Simpson 1946). Convergence usually refers to the independent appearance of similarities in distant lineages that have faced a similar environment (Arendt and Reznick 2008). In other words, convergence occurs when two species – despite initial differences between them – converge towards a shared phenotype. A classical example of convergence is the appearance of fins in fish, reptiles and mammals, all of which have evolved in aquatic environments. Parallelism refers generally to similarities in adaptation of independent populations from the same species in similar environments (Arendt and Reznick 2008). In this case, these initially identical populations diverge by the process of mutation, adaptation and genetic drift but then accumulate parallel genetic changes when subjected to the same environment. At the genetic level, the difference between convergence and parallelism initially suggested that convergence relies on different genetic mechanisms, because the species were highly diverged; in contrast, parallelism would rely on similar genetic targets due to the genetic similarities between populations. Yet, the evolution of color pattern in mammals has suggested that the difference between convergence and parallelism is not that marked at the genetic level: a single species may use different genetic paths to generate similar phenotypic changes (Steiner et al. 2007) and very distant species such as mice and bears may use

the same gene to change color (Ritland et al. 2001; Hoekstra et al. 2006).

Given that the two terms represent a somewhat false dichotomy, we subsequently use the term convergence to refer to the reproducibility of adaptation.

11.2.1 Multiple Facets of Convergence

Convergence is a term that is widely used but the associated concept is indeed quite complex. When genetic convergence is observed, it is classically considered to be strong evidence of adaptation. As it is usually very difficult to classify mutations as beneficial, neutral or deleterious, multiple independent occurrences of a mutation suggest the filtering action of natural selection (Pelosi et al. 2006), thus constituting a strong signal of adaptation. This remains the dominant usage of convergence in evolution. Yet, the existence of genetic convergence does not only reflect the action of natural selection – it also implies the lack of possible alternative beneficial mutations. That is, if many different solutions are possible, then convergence is not to be expected. Hence, the existence of convergence informs about the number of possible adaptations to an environment. These two facets of convergence – i.e., as a marker of adaptation and as an indicator of adaptive limitations – make it a powerful tool to study the structure of the adaptive landscape and the molecular bases of adaptation.

The interpretation of genetic convergence can also be complicated by the fact that it can be defined at several levels – e.g., at a specific nucleotide site, within a particular gene or even among genes that contribute to an integrated function or network (Tenaillon et al. 2012). While this appears to be just a matter of semantics, the level at which convergence is considered has consequences for inferring adaptive processes. In its strictest sense, convergence is the presence of the exact same mutation at the same nucleotide site; in this case, adaptation can be treated as a discrete genotypic space (Fig. 11.1a). However, when

mutations are pooled within genes (that may harbor several alleles), or even within functional units, the study of convergence implicitly moves toward a continuous multidimensional trait space for the adaptive process (Fig. 11.1b).

The meaning of convergence can also vary according to the time scale used to analyze it. Convergence in state occurs when the same changes are present in independent populations at the same time. In contrast, convergence in path refers to the succession of changes that is similar among populations. Hereafter we will refer to ‘state’ and ‘path’ convergence and will clarify their differences below.

11.3 Convergence and Simple Fitness Landscapes

Convergence provides insight into the shape and slope of adaptive landscapes. Adaptive landscapes were first introduced by Sewall Wright (1932) to illustrate the impact of mutations on evolutionary fitness. While they were initially defined at the genotypic level, landscapes can be extended to phenotypes. In the genetic case, with a set of n biallelic loci, the landscape is defined as a discrete hypercube (Wright 1932) (Fig. 11.1a); in the phenotypic case, the landscape can be envisioned as a continuous multidimensional space (Fisher 1930; Orr 2005) (Fig. 11.1b). Once an adaptive landscape is characterized, the dynamics of adaptation – and the path followed by populations on this landscape – can be computed.

Interactions among mutations, or epistasis, play a key role in shaping an adaptive landscape (Kauffman 1993; Gros et al. 2009). Epistasis is the influence of alleles at other sites on the effect of a mutation at a given site (Fig. 11.2). In population genetics, for two biallelic loci, epistasis can be mathematically defined as:

$$e = \log [w_{11} / (w_{10}w_{01})]$$

where w_{11} is the fitness of the genotype with a mutant allele at two different sites, w_{10} and w_{01} are the fitnesses with alternative alleles at the two

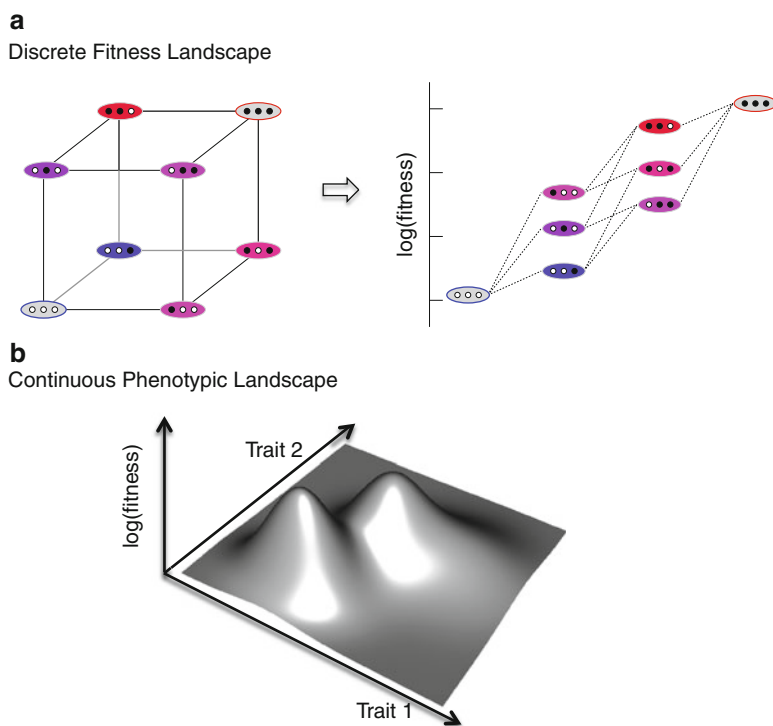


Fig. 11.1 Discrete and continuous landscapes. (a) A discrete genotypic landscape is a hypercube with as many dimensions as the number of biallelic loci. Here, three biallelic loci are arranged in a cube. Each genotype is presented with a circle for each locus, and a black or white filling to reveal the allele present. The background colors on the nodes of the hypercube are cold (i.e., blue) when

fitness is low and warm (i.e., red) when fitness is high. Another representation is given on the right panel where the height at which a node is placed depends on the log (fitness). In this example, there is no epistasis, the fitness improvements of mutations are additive in log-scale. (b) A continuous phenotypic landscape, here, with two traits

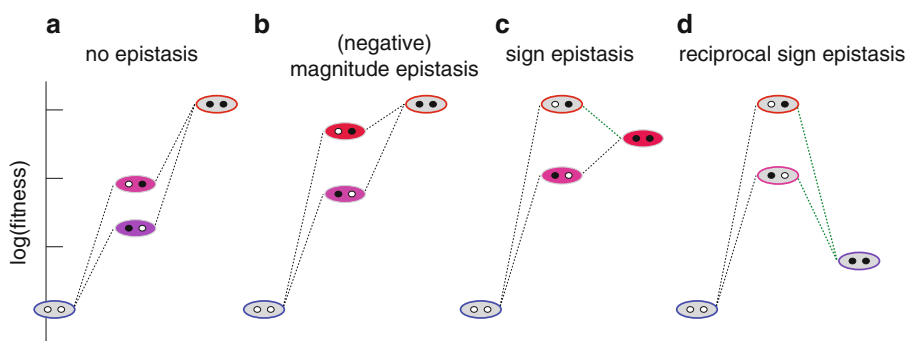


Fig. 11.2 Different types of epistasis. The different types of epistasis are presented with two biallelic loci. No epistasis is characterized by additivity of the mutation effect in the log-scale. When the order of the genotype is conserved, it is magnitude epistasis. If not, it is sign epistasis. If both mutations combined have an opposite effect, it is reciprocal sign epistasis. Reciprocal sign epistasis is a necessary but not sufficient condition for the existence

of multiple peaks. If there are several peaks, at least one must be composed of only a fraction of all the beneficial mutations that define the landscape. The fitness of such a genotype decreases with the addition of any further “beneficial” mutation. Hence, combining beneficial mutations that constitute this genotype decreases fitness, which is the definition of reciprocal sign epistasis among beneficial mutations

different sites, and w_{00} is the wild type that has, by definition, a fitness of 1.0.

In this formulation, epistasis is negative when the fitness of the double mutant (w_{11}) is lower than expected from single mutants (Fig. 11.2b–d). Positive epistasis is the opposite, when the fitness of w_{11} exceeds expectations based on the fitnesses of the single mutants. Magnitude epistasis refers to epistasis that does not affect the beneficial or deleterious status of mutations; that is, the rank of the mutant fitness can be predicted from the single mutants (in the case of beneficial mutations $w_{11} > w_{10}$, $w_{01} > 1$; Fig. 11.2b). Conversely, sign epistasis refers to mutations that are beneficial in one genotype and deleterious in the other; for example, when w_{10} and w_{01} are both > 1 but $w_{11} < w_{10}$ or $w_{11} < w_{01}$ (Fig. 11.2c). Reciprocal sign epistasis refers to conditions in which both mutations switch from beneficial to deleterious depending on the allele present at the other locus. For beneficial mutations (w_{10} , $w_{01} > 1$), this occurs when the double mutant has lower fitness than both single mutants ($w_{11} < w_{10}$ and $w_{11} < w_{01}$) (Fig. 11.2d). Epistasis is fully defined according to measure theory in term of fitness measurements (Wagner 2010), but sometimes ‘fitness’ is represented by a proxy, such as enzymatic activity or a minimum inhibitory concentration (MIC), in case of microbial resistance to a particular compound. In these latter cases, the term epistasis should be used with care because activities and MICs are at best an approximation for evolutionary fitness. However, it is worth noting that sign epistasis at the phenotypic level reflects the presence of sign epistasis at the fitness level as long as the measured phenotypes vary monotonically with fitness. This is true because a mutation either increasing or decreasing the measured phenotypes is a mutation increasing or decreasing fitness pending on the genetic background and therefore asserts the presence of sign epistasis at the fitness level. However, this parallel cannot be extended to magnitude epistasis, because it is not simple to define the null expectation of the double mutant phenotype based on the single mutant phenotypes. Should, for example, an additive or a multiplicative expectation be used? As a consequence, magnitude

epistasis should be inferred with extreme care when dealing with phenotypes rather than fitness.

11.3.1 Adaptive Landscapes in the Absence of Epistasis

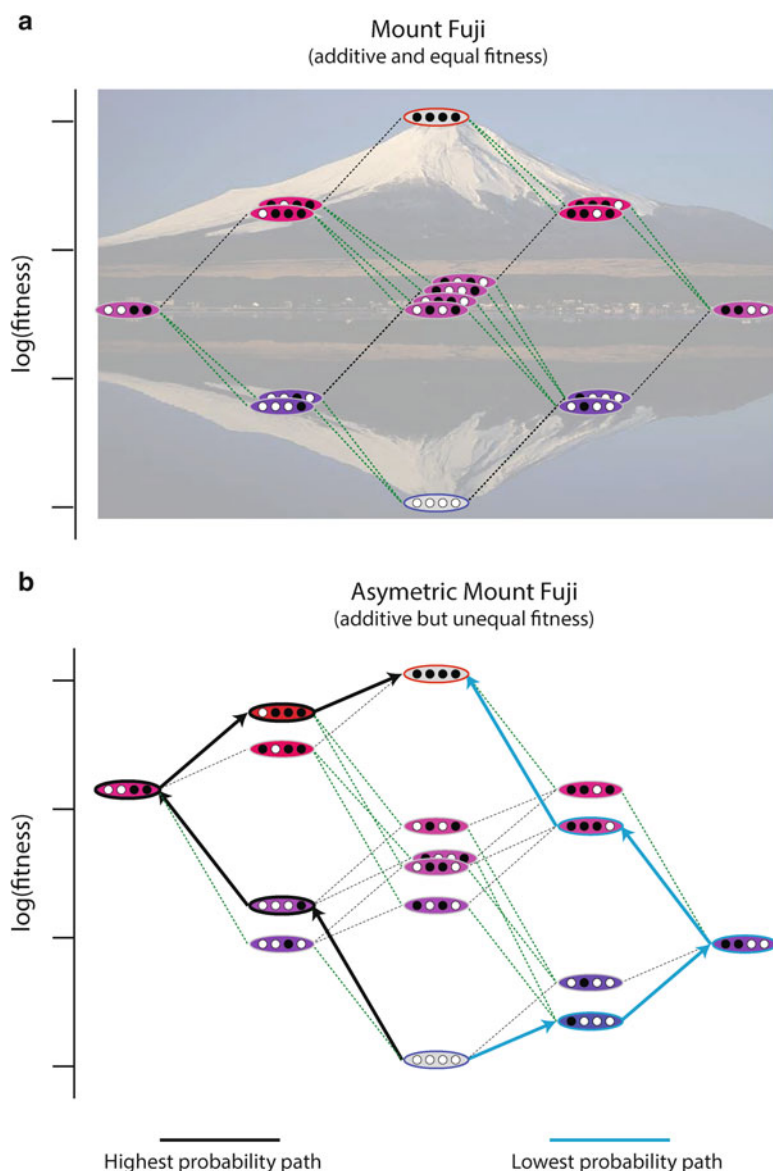
To begin to understand the relationship between epistasis, adaptive landscapes and convergence, let us first assume that there is no epistasis among beneficial mutations. In the absence of epistasis, the fitness landscape can be described by a single peak, which is somewhat reminiscent of Mount Fuji (Fig. 11.3), and whose summit is occupied by the genotype that has all of the possible beneficial mutations. In such a landscape, all populations will converge to that optimal genotype provided there is enough time. Hence, with full adaptation, state convergence will be absolute among populations while path convergence will depend on several factors (see below). Nevertheless, if many beneficial mutations are necessary to scale the peak, it may take very large amount of time to reach full adaptation.

The larger the number of mutations needed to scale the peak, the larger the number of potential adaptive paths. For N beneficial mutations there are $N!$ different paths in which mutations are added one after the other. [Even more paths are possible when reverse mutations are allowed (DePristo et al. 2007).] In the absence of epistasis all paths can be taken, although the probability of particular path is a function of mutational effects (see below), and therefore a large number of mutations will affect the level of path convergence among populations. Similarly, state convergence will diminish if populations are stopped in their adaptive process prior to reaching the fitness optimum because the populations may be on different paths and thus harbor different mutations.

If all mutations have similar fitness effects (Fig. 11.3a), then all paths are equally likely and path convergence will be poor. Yet if mutations have different fitness effects (Fig. 11.3b) some will be more likely to be selected than others, and consequently both forms of convergence will increase. With differential fitness effects, the big-effect mutations are more likely to be selected

Fig. 11.3 Path Probability under Mount Fuji Landscapes.

When mutations have additive fitness, the resulting landscape harbors a unique peak and a unique sink. **(a)** If all mutations have equal fitness effects, the highly symmetric structure of the landscape is reminiscent of Mount Fuji. All paths that lead from one genotype to another are equally likely. **(b)** When mutations are additive but vary in their fitness effects, the landscape is no more symmetric each adaptive path has a specific probability. Typically, if mutations are not limiting, the path that has highest fitness improvement at each step (the *black path* on the *left*) is associated with the highest probability and reciprocally, the one that has the lowest fitness improvement at each step (the *blue path* on the *right*) is associated with the lowest probability



first and the small-effect mutations later. If populations are small the impact of the variance in fitness effects among beneficial mutations may not be important, because mutations are limiting such that as soon as one survives drift it will be fixed. As a result, the probability of surviving genetic drift predominates over selection in small populations (Weinreich 2005). However, as the population size gets bigger, more and more beneficial mutations will be simultaneously present in

the population, and may compete in what is called 'clonal interference' (Gerrish and Lenski 1998; Desai and Fisher 2007). As the fittest mutations outcompete other mutations, both forms of convergence will be higher because mutations will be fixed approximately according to the ranks of their fitness effects. If populations get even larger, double mutants will appear, making path convergence difficult to observe. Thus, both state and path convergence are most likely to be observed

in populations that are large but not so large as to commonly house double-mutants (Paixao et al. manuscript under preparation).

11.3.2 Epistasis Affects the Likelihood of Observing Convergence

Both forms of convergence may strongly be affected by epistasis. The constraints imposed by epistasis on convergence are often referred to as ‘historical contingency’. In other words, the appearance of a mutation by chance has a strong impact on consequent steps of adaptation. Thus, epistatic interactions may strongly affect the likelihood of convergence. Moreover, epistasis can affect the adaptive landscape in several ways. It can generate a rugged fitness landscape (Poelwijk et al. 2011) or it may affect the shape and slope of a single-peaked landscape.

Epistasis impacts convergence on a single-peak fitness landscape by modifying the effect and number of beneficial mutations that are accessible to evolution. For example, in the presence of sign epistasis some mutations will be beneficial in only a subset of genetic backgrounds and therefore all the paths in which they are not beneficial will not be adaptive paths. Hence, the presence of sign epistasis in an adaptive landscape should increase both state and path convergence to some extent. Multiple adaptive peaks may result from reciprocal sign epistasis (Poelwijk et al. 2011), which is a necessary, but not sufficient, condition for the existence of multiple peaks (Fig. 11.2). Reciprocal sign epistasis is also likely to affect convergence by driving populations to diverge as they evolve on separate peaks. Hence, the reproducibility of adaptation is intimately linked to the adaptive landscape.

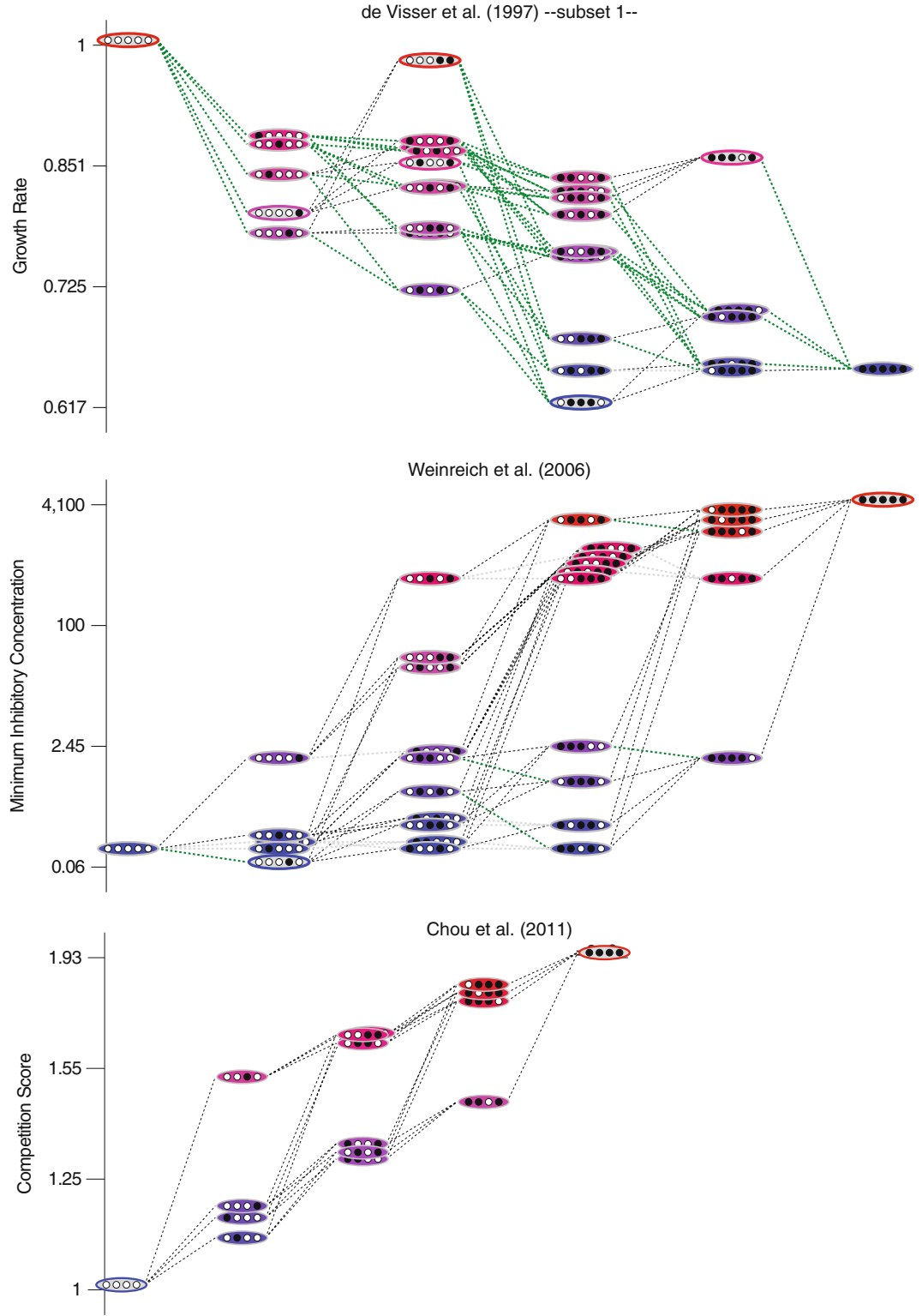
11.4 Experimental Evolution as a Tool to Study Convergence

Given the complexity of epistatic interactions and adaptive landscapes, how does one study convergence? One approach is comparative stud-

ies (Besnard et al. 2009), which may identify similar changes in distant species. Yet, when one examines the molecular details of adaptive events, it is often difficult to be sure that they resulted from the same selective pressures; that is, did the different organisms truly experience and adapt to the same environmental challenge? As a consequence, if different adaptive responses are observed among species or among populations, it is difficult to know if they result from slight differences in selective pressures or from the stochastic nature of adaptation. Alternatively, if similarities are observed, it is difficult to know if they result from the same selective pressure or appeared as collateral effects of different selective pressures. Therefore, to truly understand convergence, adaptive landscapes and the stochastic nature of the adaptive process, highly controlled environments are necessary.

Experimental evolution in the laboratory can provide such controlled conditions. Although some interesting studies have been performed with higher eukaryotes (Rose et al. 2005), we will focus on experimental evolution with microbes, mostly bacteria and yeast, with an occasional mention of viruses. The benefit of using microbes in experimental evolution comes from their short generation time, their small size, their large populations, and the ability to stop their evolution by storing them at -80°C and reviving them on demand. These properties allow population sizes of millions to billions of cells to be evolved for hundreds to thousands of generations and compared to their ancestor. Finally, as microbes have relatively small genome sizes, it has become possible to sequence complete genomes with next-generation sequencing (Herring et al. 2006; Barrick et al. 2009; Tenaillon et al. 2012).

Another advantage of experimental evolution is the ability to perform experiments in replicate. For example, the longest evolutionary experiment in the lab is the one initiated by Richard Lenski in 1988 (Lenski et al. 1991; Barrick 2009 #2), in which 12 replicate populations have been evolved for more than 50,000 generations to low nutrient conditions. We will later on refer to this experiment as the Long Term Experimental Evolution, or LTEE.



As soon as replicated experiments have been sequenced, information on the reproducibility of adaptation can be gathered at a genetic level. The pioneering work of Wichman et al. (1999) nicely illustrates the study of convergence with experimental evolution. Wichman and colleagues (1999) evolved two independent populations (one in Texas and one in Idaho) of bacteriophage phiX174, a single stranded DNA genome of 5,386 bp, and sequenced their genomes after 11 days of adaptation in a chemostat in which the phage adapted to a new host at a high temperature. They first observed that 50 % of the 14–15 nucleotides changes present in the sequenced clones from each population were identical. Hence, end-point sequencing revealed a very high level of convergence. To go further, they then followed the time frame of the appearance of each of the common mutations in each of the populations and demonstrated that the similarity of the adaptive paths was poor, because the mutations appeared in different orders. Hence, their experiment yielded different messages about convergence, depending on whether they focused on convergence in state or convergence in path.

11.4.1 Experimental Adaptive Landscapes

The study of Wichman et al. (1999) demonstrates the feasibility of studying convergence using the approach of experimental evolution. Similar approaches have been useful in inferring properties of an adaptive landscape. Generally, these studies are of one of two types. In the first, the adaptive landscape is thoroughly described for a handful

of mutations, which are then used to infer the number of possible paths on the landscape (Poelwijk et al. 2007; Kogenaru et al. 2009). In the second, empirical observations of convergence are used to infer some of the landscape's properties.

11.4.1.1 Reconstructing Adaptive Landscapes

Using site-directed mutagenesis, it is possible to reconstruct an adaptive landscape. With N observed mutations, a landscape is an N -dimensional hypercube, whose full characterization requires the construction and fitness determination of 2^N mutants (assuming two states per mutated site). To date, the experiments that have constructed adaptive landscapes have focused on four to nine mutations that were found together in an adapted genotype. Pioneering work by Lee (Lee et al. 1997), Lunzer (Lunzer et al. 2005), de Visser (de Visser et al. 1997, 2009) and Weinreich (Weinreich et al. 2006) have used this approach. However, de Visser, rather than using beneficial mutations that co-evolved together, generated a landscape from unrelated deleterious mutations with selectable phenotypes (Fig. 11.4a). Several similar studies have been published more recently (Lozovsky et al. 2009; O'Maille et al. 2008; da Silva et al. 2010; Tan et al. 2011; Khan et al. 2011; Chou et al. 2011). Many of the recent studies have focused on mutations that provide antibiotic resistance or adaptation to new culture conditions.

Interestingly, this experimental approach has revealed that the pattern of epistasis seems to vary as to whether mutations are found within a single gene or across a series of genes. Mutations within a single gene or enzyme frequently result in sign epistasis (Lee et al. 1997; Lunzer

Fig. 11.4 Three experimentally resolved adaptive landscapes. All proxies for 'fitness' values are represented here in log-scale. The landscape from de Visser et al. (1997) is a combination of deleterious mutation held in different genes of *Aspergillus niger*. Although there is a general tendency to have smaller fitness with more mutations, the landscape harbors several peaks and sinks; it is rugged. The landscape of Weinreich et al. (2006) explores the effect of mutations that have coevolved together in *E.*

coli toward a high resistance to cefotaxime. The landscape has one peak and one sink but shows pervasive sign epistasis. From the sink to the peak, only 18 direct paths are possible (out of 120). The landscape of Chou et al. (2011) reports how 4 mutations that have coevolved to allow *Methylobacterium extorquens* to grow on methanol interact together. It shows no sign epistasis and is close to a model with no epistasis

et al. 2005; Weinreich et al. 2006; O'Maille et al. 2008; Lozovsky et al. 2009; da Silva et al. 2010; Tan et al. 2011), such that mutations become beneficial conditionally on the presence of other mutations. In other words some mutations compensate the fitness deficits produced by previous mutations. For example, Weinreich (2005) observed that the β -lactamase TEM1 could improve its ability to hydrolyze the antibiotic cefotaxim by 40,000 times with a total of 5 different mutations (Fig. 11.4b). Yet, in the wild type TEM1 background only two of the individual mutations were beneficial. In this case, sign epistasis could be understood functionally. For example, the stabilizing mutation M182T had no effect in the ancestral background, but was highly beneficial once a destabilizing mutation was fixed, because the destabilizing mutations modified the active site to improve the efficiency of M182T on cefotaxim. Similarly, a mutation in the promoter was only beneficial once the enzyme started to have activity on the new substrate (Brown et al. 2009). Overall, out of the 120 different forward paths, only 18 different paths were found to be accessible due to the presence of sign epistasis.

Another interesting case of sign epistasis was observed for the 5S ribosomal RNA of the marine bacterium *Vibrio* (Lee et al. 1997). Lee et al. used the sequences of 5SrRNAs found in *V. proteolyticus* and *V. alginolyticus* that differed at four nucleotides, and constructed the 16 possible allelic combinations ($2^4 = 16$) between these two sequences. They found that only 5 out of 24 possible mutational pathways were functional, again suggesting sign epistasis between mutations. Similarly, using phylogenetic information, nine mutations in Sesquiterpene synthetase were identified as having lead to a catalytic shift in the evolution of that enzyme. Biochemical analysis of 418 combinations of the nine mutations, out of the 512 possible combinations, also revealed a rugged landscape with multiple peaks (O'Maille et al. 2008).

In the case of TEM1, the landscape had a single peak despite the high prevalence of sign epistasis. In other single-gene landscapes, multiple peaks have been found, confirming a high

frequency of sign epistasis and even the presence of reciprocal sign epistasis, which is a necessary condition for the existence of multiple peaks (Poelwijk et al. 2011). For example, in the case of DHFR gene evolution in the malaria parasite, two peaks were found when selection was due to the drug pyrimethamine and three peaks when it was to resistance to the drug antifolate chlorocycloguanil (Costanzo et al. 2011).

In contrast to single-gene studies, sign epistasis seems to be less prevalent among mutations from different genes. This conclusion is primarily based on two landmark studies. In one, Khan et al. (2011) produced all possible mutants from the first five mutations involved in the adaptation of *Escherichia coli* to the LTEE conditions. Similarly, Chou et al. (2011) built the 2^4 genotypes of the four primary mutations that permits *Methylobacterium extorquens* to grow on methanol (Fig. 11.4c). It is worth noting that in these cases, producing the combination of mutants is a much harder task than in the case of a single gene cloned on a plasmid. Moreover, while single gene studies often use a biochemical activity such as IC50 (concentration allowing 50 % of survival) or MIC as proxy for fitness, true fitness assays must be performed in the case of genome analysis.

Nonetheless, epistasis was pervasive in both of these studies. Both landscapes had a single peak, and epistasis was consistently negative in magnitude. Most mutations appeared to be beneficial in all background in Chou et al. (2011), but the magnitude of their fitness effects decreased as mutations accumulated. As a result, the authors concluded that there are 'diminishing returns' with respect to the addition of each additional new advantageous mutation. However, we suggest that this conclusion be considered with caution, as the relationship between the fitness defined in mathematical models and the measured experimental values is not necessarily trivial (Wagner 2010). In population genetics, epistasis can only be meaningful when measured as a departure of additivity on log fitness. However for historical reasons, fitness in experimental evolution is often measured as ratios of log fitness.

Once epistatic interactions are inferred experimentally, the number of paths can be studied *in silico*. To study convergence computationally, most simulations use the “low mutation, high selection” regime. In this regime, the population is mostly monomorphic; following the appearance of a mutant, the populations may instantly shift to the mutant state depending on its fitness. The *in silico* studies have shown, as mentioned previously, that the structure of the landscape will affect drastically the chance of observing convergence (Lobkovsky and Koonin 2012; Lobkovsky et al. 2011). This is also obvious from empirical studies, because one striking contrast among studies comes from the number of possible paths to the optimal genotypes. In Chou et al. (2011) all possible forward paths were possible (24 out of 24) because each represented an increase in fitness. However, among the 120 possible forward paths in the Kahn et al. (2011) landscape, only 90 were possible. Finally, only 18 (out of 120) paths led to increasing fitness in the case of β -lactamase TEM1 (Weinreich et al. 2006, Fig. 11.4).

One fundamental question is: Why are the landscapes so different? A tempting answer is that, considering a handful of beneficial mutations, there ought to be more interactions if they lie within the same gene than if they are spread throughout the genome. In a broader sense, adaptation at the genome level is likely to be possible at many different functional levels, with the different improvements less likely to interact than those within a single gene. Yet this answer is not fully satisfactory because apparently unrelated mutations in different genes can also show pervasive sign epistasis (de Visser et al. 1997). Only new and ideally larger adaptive landscapes will help to quantify the differences.

11.4.1.2 Convergence in Experimental Systems: Adaptation in Replicates

An alternative to constructing a landscape from a handful of experimentally combined mutations is replicated evolutionary experiments. Experiments of parallel evolution have been performed and interpreted in terms of convergence and the

properties of adaptive landscapes. In most studies, just like in the case of phiX174 mentioned above (Wichman et al. 1999), some shared mutations are recovered among replicates while others are specific to a single lineage (Woods et al. 2006; Herring et al. 2006; Conrad et al. 2009; Charusanti et al. 2010; Anderson et al. 2010; Toprak et al. 2012). It can be difficult to make conclusions from mutations in single lineages unless there are a large number of replicates or in the absence of a further functional characterization, because mutations in single lineages may have simply hitchhiked with beneficial mutations and therefore may not be adaptive. Yet, even these straightforward empirical comparisons shed light on the complexity of convergence that is missing from landscape reconstruction experiments. Most of these studies reveal that lineages infrequently share the same mutation but may mutate the same gene (Woods et al. 2006; Toprak et al. 2012), suggesting that the gene may be the appropriate unit to measure convergence. Taken together, these studies indicate that the unit of convergence needs to be clearly defined, because the unit confers implicit assumptions about convergence and the epistatic properties of the adaptive landscapes.

Most inferences about adaptive landscapes from replicated evolutionary experiments come either from studies with a large number of replicates or, alternatively, from studies in which a clear signal of convergence emerged with few replicates and that signal could be confirmed experimentally or with more targeted replicates. An example of the former (a large number of replicates) is represented by our recent work (Tenaillon et al. 2012), in which we adapted 115 independent replicates of *E. coli* to a low resource media with an elevated temperature. The ancestral clone to all replicates was pre-adapted for 2,000 generations to the same low resource media that was used in the LTEE. A temperature of 42.2°C was chosen for heat stress, because it was the maximal temperature that allowed a 100-fold increase in population size over a daily growth cycle. In other words, at 42.3°C the populations could continue to grow, but they would have slowly decayed to extinction

with daily 100-fold dilution to fresh medium. After 2,000 generations of adaptation at 42.2°C, we sequenced one clone of each population.

Why choose such a system? The high temperature stress was chosen because under these conditions we had no *a priori* ideas about the mutational targets that natural selection would affect; temperature affects not only the folding of every protein but also the kinetics of all reactions within the cell. Choosing a stress for which the targets of selection are known (like rifampicin resistance, for example) may bias the study of convergence. We chose an extreme environment and a reasonably long adaptive period, because we wanted both a fast response to selection and several mutations to be present in each of the sequenced clones. The low level of initial adaptation and the low resource conditions had the benefit of limiting the potential emergence of ecological interactions within each population. In rich media such interactions may occur faster than in limited resource conditions (Le Gac et al. 2008). For example, under LTEE conditions, several thousands of generations of adaptation were needed to observe ecological interactions (Le Gac et al. 2012), while only several hundred generations were necessary in richer media (Le Gac et al. 2008). Moreover strong maladaptation to the initial conditions also limits the emergence of ecological divergence, because the first steps of the adaptive process are more likely to be linked to general adaptation. While ecological interactions are interesting, they complicate analyses of adaptation and convergence and should be avoided, if possible. For example, if two morphotypes emerge and are maintained through a frequency dependent interaction (Rainey and Travisano 1998), sampling a single clone randomly from each population may generate a pattern of divergence among populations, when in fact all the replicate populations are completely convergent with respect to the emergence of two morphs.

Using the full sequences of the 115 clones, we were able to uncover several interesting facets about convergence. First, because we identified a large number of mutations, convergence could be studied at the level of individual mutations, single

genes, an operon or even on a functional level – i.e., groups of genes that contribute to the same function. As expected, the more integrated the definition of similarity, the higher was the convergence among lineages. However a substantial gain was observed when comparing convergence at the level of individual mutations (around 2 % convergence) to convergence at the gene level (>20 %) or at the functional level (>30 %). A direct consequence of this discrepancy was that we could identify about 30 functional units that were affected repeatedly by adaptive mutations, but that the number of mutations affecting those functional units was extremely large. In our large study, we observed several hundred different beneficial sites, but simple extrapolations suggested that several thousands of potentially beneficial sites are present in the genome. It is important to note that the large diversity of mutations affecting a gene or function was not only restricted to the inactivation of genes but also extended to essential genes such as the RNA polymerase or the genes involved in the bacterial rod shape. Within these genes we could estimate that tens to hundreds of different mutations may have adaptive potential.

Second, through statistical analysis of the association of these mutations among lineages, we were able to show that epistasis was pervasive. Alternative mutations within a gene or a functional unit appeared to be strongly negative, because combinations of mutations within a gene were rarely found. This observation has an intuitive explanation within a functional unit: if two different mutations have the same phenotypic effect, once one is selected for, the second is likely to be useless, if not harmful.

However, evidence of epistasis was not limited to within gene interactions; epistasis among mutations across genes and functional units was also strong enough to leave a statistical signal. For example, the presence of mutations in *rpoD*, the capsule operons, or the genes involved in cell shape were conditional on the presence of mutations in the RNA polymerase β or β' subunits. This suggests some positive epistasis among these genes and their functions. Altogether, the pattern of these interactions suggested

two alternatives adaptive trajectories and fitness peaks (in a broad sense). One consisted of mutations in the RNA polymerase β - β' subunit, and the other consisted of mutations in the *rho* factor gene, with the two trajectories being partially exclusive. Interestingly a systemic analysis of the impact of a mutation in *rho* on the fitness effect of transposon-based gene inactivation (Freddolino et al. 2012) revealed that a single mutation in that gene was enough to drastically change the pattern of selection over the whole genome. This confirms the idea that affecting global regulators such as *rho*, which is a termination factor, or RNA polymerase can have a large epistatic impact by making some previously beneficial mutations deleterious and by, reciprocally, making some previously deleterious mutations beneficial.

11.4.1.3 Evidence of Multiple Peaks

Our large-scale experiment is not alone in suggesting the existence of multiple fitness peaks. Several other evolution experiments with a lower number of replicates have also suggested the existence of multiple peaks (Burch and Chao 2000; Kvitek and Sherlock 2011). In these experiments, a fewer number of replicates were sufficient to suggest adaptation by competing pathways that were subsequently validated experimentally.

For example, a two-peaked adaptive landscape was inferred from an experiment with two phage populations of $\phi 6$ (Burch and Chao 2000). Burch and Chao (2000) used two clones that accumulated deleterious mutations and were allowed to recover through the accumulation of beneficial mutations (Burch and Chao 1999). These founder clones (clone A and clone B) were used to find five new replicate populations and were evolved for 100 generations. Derived populations from clone A evolved to higher fitness, while derived populations from clone B evolved to lower fitness, suggesting that clones A and B resided on different fitness peaks.

In a study of yeast adaptation to glucose limitation, Kvitek and Sherlock (2011) also found evidence for two competing adaptive genetic paths. They found that glucose transport into the cell could be improved by two means – either inactivation of a negative regulator of the glucose

sensing signal transduction pathway or by an amplification of genes encoding a high affinity glucose transporter. While individual mutations were highly beneficial, they were deleterious in combination. Based on these observations, the glucose transporter experiments appear to reveal a case of negative epistasis within a functional unit because both mutations increase hexose transporter transcription.

Interestingly one can find evidence of multiple peaks even at the gene level. Using the same enzyme and selective pressures as Weinreich (Weinreich et al. 2006), Salverda et al. (2011) adapted β -lactamase TEM1 to cefotaxim by random mutagenesis. While many lines adapted with the mutations studied by Weinreich, many alternative mutations also appeared to be beneficial. Moreover, while Weinreich suggested a single peak, in this analysis they found that populations could move toward different optima, depending on the initial mutation that was fixed.

11.4.1.4 Reduced Landscapes Versus Replicate Adaptation

There are some discrepancies between the different approaches. Thorough analysis of a handful of beneficial mutations suggests a single peak, while adaptation experiments commonly suggest multiple genetic solutions based on alternative competing genetic pathways. Even in the case of a single gene we have a contrast between a single and multiple peaks for the same enzyme. What may explain such discrepancies?

The first difference between these approaches may be due to sampling biases in adaptive landscape reconstruction. Typically the mutations chosen to study were found in combination in a single adapted population. This may drastically limit the chance to find multiple adaptive peaks. For instance in the case of the LTEE, the landscape had a single peak (Khan et al. 2011). Yet that landscape included a *spoT* mutation and it has been shown that among the replicates many alternative alleles of *spoT* were recovered (Ostrowski et al. 2008). Had these alleles been included in the landscape reconstruction, a different picture may have emerged (Fig. 11.5b). Indeed the only landscape reconstruction that

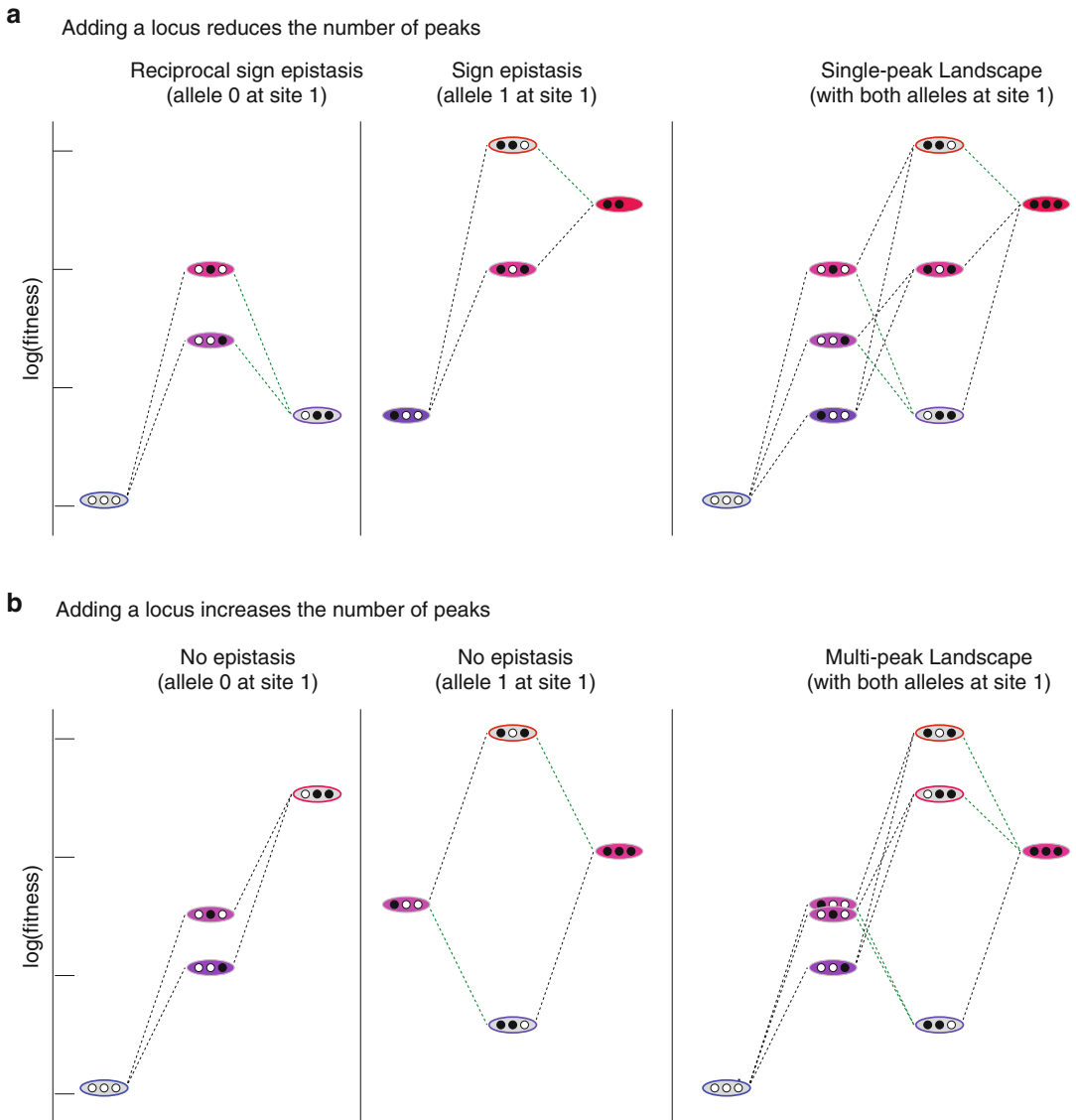


Fig. 11.5 How landscapes change with more loci. The addition/discovery of a new polymorphic locus can change the number of peaks in the landscape. **(a)** Although loci 2 and 3 show (reciprocal) sign epistasis, taking into account the two alleles in locus 1 leads to a single peak

landscape. **(b)** When the allele at locus 1 is fixed, loci 2 and 3 show no epistasis (but their fitness rank depends on the allele at locus 1). When the locus 1 is included in the landscape, the landscape harbors two peaks, a strong reciprocal sign epistasis

included mutations on a genome scale that were not isolated in combination discovered several peaks (de Visser et al. 1997).

For technical reasons it is difficult to generate a full landscape for many sites. The number of mutants to be constructed and characterized increase by 2^N . The biggest such landscape recon-

structed thus far has been with 9 mutations, and it could not be fully reconstructed because only 418 of 512 combinations were achievable (O'Maille et al. 2008). Yet, adaptation at the genome level or even at the gene level reveals that tens to thousands of mutations may be recruited in the adaptive process. Thus, complete reconstruction

of the landscape is beyond reach with current technology. Yet, as we have noted, restricting the analysis of convergence to a fraction of the adaptive landscape may be misleading. Adding more sites and loci may not only decrease path convergence as alternative paths are added, but it may also change the ruggedness of the landscape as peaks disappear or appear with an increasing number of loci. As we mentioned in the case of *spoT* alleles, adding a site may add new peaks, but it could also reduce the number of peaks. A landscape made of two mutations with reciprocal sign epistasis suggests a double-peaked fitness landscape. Yet adding a third site may result in a single peak (Fig. 11.5a).

11.4.1.5 Reducing the Complexity of Discrete Adaptive Landscape

As we mentioned previously, a fitness landscape is a discrete hypercube with a set of n biallelic loci. As evolution experiments increase their replication and the ability to detect genetic variants, this hypercube quickly becomes intractable. To reduce the dimensionality of the hypercube to be analyzed it may be tempting to aggregate mutations according to their function. For instance the different alleles of *spoT* in the LTEE could be all considered similar. Going even further all the loci affecting the same functional units could be aggregated into a single locus with two alleles. In the case of thermal adaptation, for instance, the various alleles affecting the six genes that contribute to bacterial shape are never found in combination; thus they can be considered as single mutation at a bi-allelic locus, and the mutation rate toward the mutated allele can be defined as the rate of production of any of the underlying mutations. Analyzing convergence on the resulting landscape clearly means a shift in the way we study convergence. Rather than genetic convergence, we move towards a vision of convergence at a phenotypic level. This is not shocking, as it is through their phenotypic effects that genotypes are selected. Reverting to the Gould quote, the original question was not whether self-conscious organism with DNA identical to ours would emerge with a replay of the

tape of life but rather whether a phenotype – self-consciousness – would inevitably emerge.

Yet, to be able to aggregate mutations affecting a gene or a functional unit, mutations need to have similar fitness and phenotypic effects. This is often not the case. First, mutations in the same gene may have different selective effects (Rodriguez-Verdugo et al. 2013). Even in the case of a gene inactivation, an early or late stop codon, a frameshift or a full gene deletion may have different fitness effects. Moreover, the phenotypes of the different mutations may differ. In the LTEE, it was shown that alternative *SpoT* selected alleles had different side effects such as different fitness effects in other environments (Ostrowski et al. 2005, 2008). In the thermal adaptation experiment, most lineages adapted through changes in the RNA polymerase. However, among these changes 10 % provided resistance to the antibiotic rifampicin (Rodriguez-Verdugo et al. 2013). This means that depending on the allele chosen, populations may fully survive an antibiotic treatment or be eradicated. This illustrates how drastic the collateral effects of different alleles may be. This observation is not restricted to bacteria, as parallel adaptation of yeast to fungicide nystatin also leads to diversification among lineages on alternative environments (Gerstein et al. 2012).

Rather than considering all mutations in a gene or functional unit to be a single allele, it may be more realistic to consider them as alternative alleles at single locus. This would reduce the dimensionality of the landscape to $n = \prod_L n_i$, where n is the number of genotypes, L the number of loci and n_i the number of allele per loci. This formalism allows the specificity of alleles to be taken into account, and reduces the number of genotypes. However, this simplification may be limiting as sometimes multiple mutations may arise within a gene. This was the case in RNA polymerase in the thermal adaptation experiment and in the rod shape genes in the LTEE (Philippe et al. 2009). In addition, the number of genotypes to be characterized remains extremely large and suggests that this formalism provides only a modest improvement over the hypercube approach.

Finally, integration of convergence at the gene or functional level based on the presence of mutations could also be misleading not only about their side effects but also in their primary phenotypic action. For example, the same genes have been target for mutations in different environmental conditions. In several experiments, mutations in the RNA polymerase have been found. Two of these studies have enough replicates to perform some comparative statistical analysis. Conrad et al. (2010), evolved 45 lineages of *E. coli* to glycerol minimal media and found that 37 had a mutation in the *rpoBC* complex. In our thermal adaptation experiment (Tenaillon et al. 2012), 91 lineages out of 114 had mutations in one of these two genes. One could easily conclude there is functional convergence between these two experiments. However, the patterns of mutations were quite distinct. Out of the 37 mutants found in *rpoBC* in the glycerol adapted lineages, two were in *rpoB* and 35 in *rpoC*. All of the *rpoC* mutants were deletions, 31 of them being an identical 9 bp deletion. In the thermal adaptation experiment, 75 lines had a mutation in *rpoB*, 21 lineages had a mutation in *rpoC* and five lineages had a mutation in both. Of these, one of the 21 *rpoC* mutations was a 3 bp deletion in a distinct region from the deletions observed by Conrad et al. (2010). Hence, the ratio of mutations among the two genes (Fisher test (FT) $p < 10^{-13}$), the type of mutations within *rpoC* (FT, $p < 10^{-13}$) and the location of mutations are so different between experiments that it strongly suggests that the mutations must have different phenotypic effects despite being in the same genes.

11.4.1.6 A Continuous Approach

Are there any fruitful alternatives to these discrete, small-scale attempts to reconstruct an adaptive landscape? Over the last decade, a phenomenological continuous model, the Fisher Geometric model of adaptation, has retained the attention of the scientific community. In his book “The Genetical Theory of Natural Selection”, R.A. Fisher (1930) outlined a model in which an organism is characterized by a set of independent phenotypes, each corresponding to an axis in an

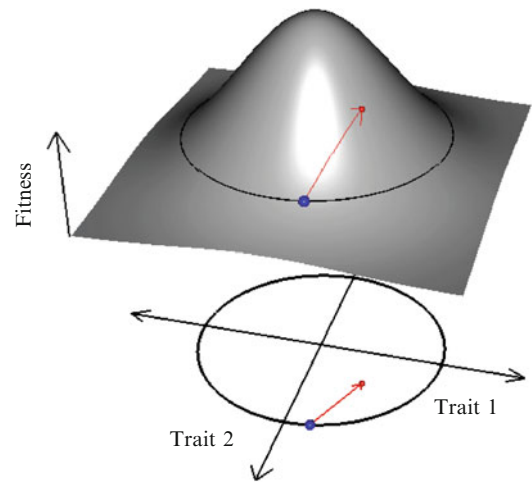


Fig. 11.6 Fisher's Geometric model of adaptation.

Fisher's geometric model is an idealized model of phenotypic landscape. In this model, an organism is defined by a number of independent phenotypes (here 2). The number of phenotypes is what is called phenotypic complexity. The model assumes the existence of an optimal combination of phenotype having maximal fitness and therefore the existence of a single peak. The more organisms are distant from that optimal combination, the lower are their fitnesses. Here an organism presented in blue, mutated to a new combination of phenotypes presented in red. Under the 3D presentation, a 2D vision of the phenotypic change is presented

n-dimensional vector space. An individual, with its specific phenotypic values, is represented by a point in this space and a mutation as a vector (Fig. 11.6). All phenotypes are supposed to be under stabilizing selection, i.e., there is an optimal value for each phenotype. As a consequence, an optimal combination of phenotypes exists and is defined to be the center of the space. The distance to that optimum determines the fitness of individuals. Several functions can be used to describe how fitness declines with the distance to the optimum. For the sake of similarity with multi-locus models, this function has been selected to be of the form.

$$W(d) = e^{-d^Q}$$

where d is the distance to the optimum and Q is an epistasis parameter (Tenaillon et al. 2007; Gros et al. 2009; Gros and Tenaillon 2009).

This model requires only three meta-parameters:

- The number of phenotypic dimensions (or the phenotypic complexity).
- The fitness function, which represents the decay of fitness as a function of the distance to the optimum in the simplest case but that can also be extended to include some asymmetry.
- The way a mutation affects an organism. Here again, the simplest model will assume isotropy of mutations (mutation directions is random), but some forms of anisotropy (some preferential directions for mutations exist) can be added.

Despite its high level of abstraction, this model has been very valuable for describing various properties of the adaptive landscapes revealed by experimental evolution. This model has shed light into the distribution of mutational effects, epistatic interactions, fitness trajectories and even the existence of fitness equilibria depending on population sizes (Hartl and Taubes 1996, 1998; Burch and Chao 1999; Orr 1998, 2000, 2005, 2006; Martin and Lenormand 2008, 2006a, b; Martin et al. 2007; Poon and Otto 2000; Waxman and Welch 2005; Welch and Waxman 2003; Wang et al. 2010; Silander et al. 2007; Lourenco et al. 2011; Chevin et al. 2010; Gros et al. 2009; Gros and Tenaillon 2009; Tenaillon et al. 2007; Le Nagard et al. 2011; Trindade et al. 2012; Sousa et al. 2012). For instance, if the number of phenotypic dimensions is n , the log-fitness of populations having an effective size of N_e will converge towards an equilibrium value of $-n/(2Q N_e)$. In other words, when complexity is high, it is difficult to optimize all the traits simultaneously and a high population size is required for the population to reach a high fitness.

It is unclear why this simple model seems to be a reasonable approximation, and indeed few authors have addressed this question. However, one apparent limitation of the model is its continuity. As long as the vision of adaptation was dominated by the idea that there were only a very few potential beneficial mutations, the idea of a continuum seemed inaccurate; discrete landscapes appeared to be the best description of the adaptive landscape with only a few beneficial

mutations. However, we have shown that in many environments at the gene or genome level the number of beneficial mutations is large – so large, in fact, that it prevents reconstruction of discrete landscapes. Even within highly constrained essential gene in the *E. coli* genome, a large diversity of alleles with slightly different fitness effects and phenotypic effects appear to be beneficial (Tenaillon et al. 2012). This suggests that a continuous phenotypic space is appropriate and may help explain some of the success of the model.

Yet, Fisher's model of phenotypic evolution does not explicitly suppose an underlying genetic model. In its classical form, mutations within the model are assumed to affect all traits in what is called 'universal pleiotropy', pleiotropy referring to the action of a single mutation on several phenotypes (Wagner and Zhang 2011). However, loci with different levels of pleiotropy and varying mutational spectra may be included in the mutation process (Chevin et al. 2010; Lourenco et al. 2011). This considerably limits the mathematical tractability of the model but allows analyses that may lead to a better understanding of the factors that explain convergence at the gene level (Chevin et al. 2010; Lourenco et al. 2011). In fact, restricted pleiotropy appears to be an important determinant of convergence. If all genes can affect all phenotypes, then many alternative solutions exist; conversely if a gene affects only a subset of phenotypes, then some convergence at the genetic level may be observed.

A coupling between Fisher's model and more explicit genetic models will be required to go beyond these observations. There are several ways to do so. The first one, which is quite similar to the approach of Chevin et al. (2010), is to define genetic loci with precise effects in the phenotypic space. Rather than doing so at the gene level only, defining a finite number of possible mutations within each gene may allow a more precise and multi-scale analysis of convergence. The second approach is to define a realistic and complex genetic system, based for instance on mechanistic properties of real organisms (e.g., regulatory or metabolic networks). The aim is then to translate the properties of this model in terms of Fisher model and to study convergence in this

defined space. We have studied, for example, adaptive neural network mimicking signalization pathways (Le Nagard et al. 2011) and found that the adaptive properties of the model could be understood through the prism of Fisher's model.

11.4.1.7 Phenotypic Convergence

If a continuous approach may help to better describe adaptive landscapes, it makes the analysis of convergence more difficult because there is no such thing as whole-organism phenotyping. Thus, ideally the identification of mutations at a genetic level should be sufficient to uncover the phenotypes that are under selection. This may be easily achievable for the case of a well-known gene inactivated through stop or frameshift mutations, but in most cases genetic experiments have to be performed to infer the phenotypic consequences of mutations (Philippe et al. 2009; Conrad et al. 2009; Applebee et al. 2011). Part of this is due to the fact many mutations involved in the first steps of adaptation affect global regulators that are highly pleiotropic (Hindre et al. 2012). Unraveling the phenotypic effect of these mutation can be done, but may require several years of investigation (Conrad et al. 2010).

A few options have been used to study convergence at a phenotypic level. One initial strategy is to focus on a phenotype that is known to be under tight selection under the conditions of the experiment. For example, Reynolds (2000) studied the phenotypic effect of compensatory mutations in the RNA polymerase following the acquisition of a mutation providing rifampicin resistance. She convincingly showed that different mutants improved the efficiency of transcription that had been deteriorated by the resistance mutation. This kind of experiment is interesting but can only be applied to genes that have been identified as a target of adaptation. However, as only a specific phenotype is studied, this approach may hinder the understanding of collateral phenotypic effects and therefore provides only a partial vision of convergence.

Several phenotypes can also be studied simultaneously. Convergence or divergence is then analyzed in term of movements in a multidimensional space that can be visualized with principal

component analysis or multidimensional scaling. Populations evolving in similar directions in that space will be said to converge. Tyerman et al. (2008) used different statistical properties of growth curves to infer convergence using principal component analysis. To have an even broader vision, “-omics” approaches can be used (Cooper et al. 2003; Fong et al. 2005). Cooper et al. (2003) used transcriptomic characterization of LTEE lineages to uncover some parallel changes in two different lineages. They showed that 59 genes had their regulation affected in similar ways in the two lineages and could retrieve the signature of a global regulator, *spoT*.

The most integrated analysis of convergence comes from the work of Lewis et al. (2010), in which they used proteomics and transcriptomics in concert with a genome scale metabolic network. Though their analysis does not study convergence between lines per se, it analyses the similarity between the phenotypic evolution of adapted lineages and the predicted optimal flux in metabolic networks. This approach suggests convincingly that the phenotypes of the evolved lineages converge quantitatively towards the predicted optimality, and that this convergence occurs despite differences at the genetic level. This appears to be a very promising line of research, though it is currently limited to specific adaptations dealing with metabolic changes.

11.5 Conclusion

Throughout this chapter we have shown that all the different facets of convergence and the reproducibility of adaptation can be thoroughly studied with experimental evolution. Reductionist approaches, in which a handful of beneficial mutations are studied, have shed light on the importance of epistasis in shaping the adaptive landscape and adaptive path. Some of the limitations of this approach have been overcome by studying the large diversity of beneficial mutations recovered in parallel adaption experiments by the application of whole genome sequencing.

Connecting discrete genetic models and continuous trait models may be an interesting venue

for future research. Indeed, the ultimate goal is to study convergence among genotypes, phenotypes and fitness and to assess the parameter space in which all three approaches provide a coherent picture. The fitness landscape maps genotype to fitness, Fisher's model relates phenotypes to fitness maps, and quantitative genetics study the mapping of genotype to phenotype. The most promising way to combine these three mappings will presumably be with systems biology approaches that integrate the predictions of mechanistic models with the genotypes, phenotypes and fitness improvements of evolved populations (Lewis et al. 2010).

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